Business Requirements & Commercial Proposal

WhatsApp-Based Valet Parking Management SaaS Platform

Version 1.0 | July 2026
Confidential – Product Planning & Commercial Evaluation

1. Executive Summary

This proposal defines the business and product requirements for a multi-tenant, WhatsApp-first valet parking management platform offered as a centrally operated SaaS service. Restaurants, hotels, event venues, clubs and similar establishments will subscribe to the platform rather than purchase or host the software. Each customer will onboard its own WhatsApp Business number and use a web application for valet and management operations.

The platform replaces paper valet tokens and fragmented phone-based coordination with a digital workflow: a guest scans a venue-specific QR code, initiates a WhatsApp conversation, submits vehicle details, receives parking updates, and later requests vehicle retrieval through WhatsApp. Valet staff receive and process requests through a real-time mobile-friendly web application.

The commercial objective is to create a scalable recurring-revenue SaaS product with low marginal infrastructure cost, strong tenant isolation, and a simple onboarding process for new establishments.

2. Business Problem

- Paper tokens can be lost, damaged, duplicated or difficult to reconcile.
- Guests often need to physically approach a valet desk and wait before a retrieval request is initiated.
- Valet teams may lack a single real-time view of arrivals, parked vehicles, parking locations, key tags and retrieval queues.
- Managers have limited visibility into wait times, staff productivity, peak periods and service history.
- Manual processes make accountability, audit history and issue investigation difficult.
- Existing app-based solutions may create friction by requiring guests to install an application.

3. Proposed Solution

A guest-facing WhatsApp experience combined with a multi-tenant web application for valets, managers and platform administrators. Guests require no new application or account. Each venue displays a unique QR code that opens the venue's onboarded WhatsApp Business conversation with context identifying the venue or entry point.

Actor	Primary Interface	Core Capabilities
Guest	WhatsApp	Start valet request, provide vehicle details, receive status, request car retrieval.
Valet	Responsive Web App / PWA	Accept jobs, collect vehicle, record parking/key location, retrieve and deliver vehicle.
Venue Manager	Web Dashboard	Monitor queues, vehicles, valets, wait times, search records and review operations.
Customer Admin	Web Dashboard	Manage branches/venues, users, valets, QR codes and access.
Platform Admin	Central SaaS Admin	Onboard tenants, administer subscriptions, support customers and monitor platform health.

4. Commercial Operating Model

The product will operate as one centrally managed SaaS platform. Customers do not receive source code or separate application deployments. Multiple customers share the application and underlying infrastructure while data is logically isolated using tenant identifiers and authorization controls.

- Each restaurant group or subscribing business is a tenant/account.
- A tenant may operate one or more venues/branches.
- Each tenant connects or onboards its own WhatsApp Business number through the supported Meta onboarding process.
- Subscription pricing should preferably be charged per venue/location, with optional volume tiers or enterprise agreements.
- Applicable WhatsApp/Meta messaging charges should be contractually treated as customer pass-through costs or billed directly to the customer's WhatsApp Business Account wherever the onboarding model permits.
- Premium modules can later include advanced analytics, SLA reporting, integrations, branding, vehicle-condition photos and enterprise support.

5. Target Market

- Restaurants and restaurant chains with valet parking.
- Hotels, resorts and banquet facilities.
- Weddings, convention centres and event venues.

- Clubs and premium entertainment venues.
- Hospitals and commercial establishments offering managed valet services.
- Third-party valet operators as a future customer segment.

6. Business Objectives & Success Metrics

Objective	Illustrative KPI
Reduce guest retrieval friction	Guest can request a car remotely via WhatsApp.
Improve operational visibility	100% of active valet sessions visible in real time.
Improve retrieval performance	Track median and P90 request-to-ready time.
Increase accountability	Auditable event history for each valet session.
Enable scalable commercialization	New venue onboarded without separate deployment/database.
Maintain reliability	Production availability target defined by commercial tier; initial target ≥99.5%.
Drive adoption	High percentage of valet sessions initiated digitally versus manual fallback.

7. Scope – MVP / Commercial Pilot

- Multi-tenant customer and venue management.
- Customer-specific WhatsApp Business number onboarding/integration.
- Venue and entrance-specific QR code generation.
- Guest WhatsApp conversational workflow.
- Vehicle registration number capture and normalization.
- Valet request creation and real-time queue.
- Atomic valet job acceptance to prevent duplicate assignment.
- Vehicle lifecycle state management.
- Parking zone/slot and key-tag tracking.
- Vehicle retrieval request through WhatsApp.
- Valet mobile-responsive PWA.
- Manager operations dashboard and vehicle search.
- Role-based access control and tenant isolation.
- Session event/audit history.
- Basic operational reporting.
- Platform administration for customer onboarding and support.

8. End-to-End Business Workflow

Arrival: Guest scans a venue QR → WhatsApp opens with venue context → guest initiates conversation → system requests/validates vehicle registration → parking request is created →

valet accepts → vehicle is collected → valet records parking location/key tag → guest receives confirmation.

Retrieval: Guest selects or sends 'Get My Car' → active session is identified using the WhatsApp identity and venue context → retrieval request enters the real-time queue → valet accepts → vehicle is retrieved → status updates are sent → vehicle is delivered → session is completed and retained for audit/reporting.

9. Functional Requirements

ID	Requirement	Priority
FR-01	Platform shall support multiple isolated tenants and multiple venues per tenant.	Must
FR-02	Platform shall allow each customer to connect its supported WhatsApp Business identity.	Must
FR-03	Platform shall generate unique QR entry points by venue and optionally entrance/event.	Must
FR-04	Guest shall be able to initiate and complete the normal valet journey without installing an app.	Must
FR-05	System shall create one active valet session linked to tenant, venue, guest and vehicle.	Must
FR-06	Valets shall see real-time pickup and retrieval queues for authorized venues.	Must
FR-07	Only one valet shall be able to accept a given job at a time.	Must
FR-08	Valet shall record parking zone/slot and key tag before marking a vehicle parked.	Must
FR-09	Guest shall be able to request retrieval from WhatsApp.	Must
FR-10	Managers shall search by vehicle, request/session identifier and relevant operational fields.	Must
FR-11	System shall retain timestamped lifecycle events	Must

	and actor information.	
FR-12	Administrators shall manage users, roles, venues, valets and QR codes.	Must
FR-13	System should provide operational metrics and wait-time reporting.	Should
FR-14	System should support optional vehicle-condition photos in a later release.	Should
FR-15	System may support billing/subscription automation in a later commercial release.	Could

10. Roles and Access

Role	Access
Platform Super Admin	All tenants for platform administration; tightly controlled and audited.
Tenant Admin	Customer account, all owned venues, users and configuration.
Venue Manager	Assigned venue operations, reports and staff.
Valet	Operational queues and assigned/accepted jobs at authorized venues.
Guest	Only the WhatsApp interaction associated with their valet session.

Tenant context must be derived from authenticated identity and server-side configuration, not trusted from arbitrary client-supplied parameters. Database Row Level Security may be used as defense in depth where supported.

11. Core Data Model

Recommended entities include Tenant/Account, Venue, User, Role, User-Venue Access, Valet, Guest, Vehicle, Parking Zone, Parking Slot, QR Code, Valet Session, Session Event, WhatsApp Account/Configuration and Subscription. Core operational records should carry tenant_id and venue_id as appropriate. The Valet Session should be governed by an explicit state machine such as REQUESTED → ASSIGNED → VEHICLE_COLLECTED → PARKED → RETRIEVAL_REQUESTED → RETRIEVING → READY → DELIVERED/COMPLETED, with controlled exception states.

12. Non-Functional Requirements

- Security: HTTPS, secure secrets management, WhatsApp webhook verification, RBAC, tenant isolation, audit logging and least-privilege access.

- Privacy: collect only information required for valet operations; establish retention/deletion policies for phone numbers, vehicle identifiers and photos.
- Performance: routine API actions should target sub-second server response under normal load; queue updates should appear near real time.
- Scalability: architecture must add tenants without separate deployments and scale based on aggregate traffic.
- Availability: production components should use paid, always-on services once commercial SLAs are offered.
- Resilience: database backups, restoration procedure, retry/idempotency for webhooks, and graceful handling of WhatsApp or network outages.
- Observability: structured logs, error tracking, uptime monitoring and operational alerts.
- Mobile usability: valet workflow must work effectively on common mobile browsers and poor-to-moderate networks.

13. Recommended Technical Architecture

Initial implementation should be a modular monolith to minimize cost and operational complexity: React/TypeScript responsive frontend/PWA; Python FastAPI backend; PostgreSQL database; SQLAlchemy/Alembic; managed authentication; realtime updates using a managed realtime capability or WebSockets; object storage for optional images; and direct integration with the official WhatsApp Business Platform. A managed PostgreSQL platform such as Supabase can accelerate the pilot, while the design should avoid hard dependency on proprietary features where practical.

One production application and database can serve all tenants. Dedicated infrastructure should be offered only when justified by enterprise contractual, regulatory, isolation or scale requirements.

14. WhatsApp Integration & Cost Principles

Each customer should onboard its own WhatsApp Business number/account into the SaaS-supported integration model. The guest initiates the conversation by scanning a QR and sending a message. The product should keep the normal valet journey within the customer-service conversation window wherever permitted, minimizing chargeable outbound template usage. Pricing and category rules are controlled by Meta and can change; therefore commercial contracts should not promise permanently free WhatsApp messaging.

- Track WhatsApp usage by tenant for transparency and reconciliation.
- Keep operational notifications service/utility-focused; do not mix marketing into the core valet workflow.
- Use approved templates when required outside the permitted service window.
- Define whether Meta charges are billed directly to customers or passed through separately from the SaaS subscription.

- Do not include unlimited third-party messaging costs in a fixed subscription without fair-use protections.

15. Indicative Commercial Pricing Strategy

The following is a starting hypothesis for market testing, not a final price commitment:

Plan	Indicative Price	Positioning
Pilot	Free or ₹0–₹999 for 30–60 days	One venue; prove operational value and collect feedback.
Standard	₹2,499–₹3,999 / venue / month	Core WhatsApp valet workflow, PWA, manager dashboard and standard reporting.
Pro	₹5,999–₹9,999 / venue / month	Advanced analytics, branding, richer reports, photos and priority support.
Enterprise	Custom	Chains, centralized dashboards, SSO/integrations, negotiated SLA and dedicated options.

Final pricing should be validated against the restaurant's valet volume, current manual cost, reduction in guest wait/friction, operational savings and willingness to pay. Annual prepayment discounts and one-time onboarding/setup fees may improve cash flow. GST and applicable taxes should be handled separately.

16. Indicative Cost Structure

For development and a non-critical pilot, free tiers can minimize spend. For commercial production, budget for a paid managed database, always-on backend compute, domain, monitoring and backups. A lean early production platform may begin in the low-thousands of rupees per month for shared infrastructure, but actual cost depends on traffic, storage, provider pricing and support requirements. WhatsApp charges are variable and should be separately accounted for. Infrastructure costs are shared across tenants, producing favorable unit economics as customer count grows.

All monetary figures in this proposal are planning estimates and must be revalidated against current provider price lists before launch or contractual commitments.

17. Delivery Roadmap

Phase	Duration	Deliverables
0 – Discovery & UX	3–5 days	Finalize workflow, tenant model, states, wireframes and pilot success criteria.

1 – Foundation	1 week	Repository, environments, database schema, authentication, tenancy and RBAC.
2 – Core Operations	1–2 weeks	Valet session APIs, state machine, queues, PWA and audit events.
3 – WhatsApp & QR	1 week	Customer WhatsApp integration, webhooks, conversation flow and QR generation.
4 – Management	1 week	Manager dashboard, search, user/valet administration and basic metrics.
5 – Pilot Hardening	1–2 weeks	Security, idempotency, monitoring, load tests, backups and operational fallback.
6 – Commercialization	Post-pilot	Subscriptions, onboarding automation, terms/privacy, support process and analytics.

18. Pilot Plan

- Select one restaurant with meaningful valet volume and an engaged operations manager.
- Run a controlled soft launch alongside the existing manual process.
- Train valets and managers with a short operational guide.
- Measure number of sessions, QR adoption, pickup/retrieval times, failed/abandoned flows, manual interventions and user feedback.
- Test peak departure bursts, duplicate requests, phone/network failures and manual fallback.
- Complete a 30–60 day review before broad commercial rollout.

19. Key Risks and Mitigations

Risk	Mitigation
WhatsApp policy/pricing changes	Use official APIs; isolate messaging layer; pass through variable charges; review policies regularly.
Guest does not use WhatsApp	Provide a manager/valet-created manual session fallback.
Internet outage at venue	Design retryable actions, visible sync state and documented fallback process.
Cross-tenant data exposure	Server-enforced tenant context, RBAC, RLS defense in depth and automated security tests.

Duplicate valet acceptance	Atomic database transition/locking and idempotent APIs.
Peak retrieval surge	Prioritized queue, wait-time visibility, load testing and later smart assignment.
Incorrect vehicle/key mapping	Required confirmation steps, searchable identifiers and immutable event history.
Low staff adoption	Minimal-tap PWA workflow, training and pilot feedback iterations.

20. Out of Scope for Initial MVP

Native iOS/Android apps, AI chatbot, license-plate recognition, GPS tracking, complex parking optimization, automated valet dispatch, loyalty/marketing campaigns, payment gateway, microservices, Kafka/Kubernetes and enterprise integrations are excluded from the initial MVP unless required by the pilot.

21. Commercial Readiness Requirements

- Terms of Service and customer agreement.
- Privacy policy and data-processing terms appropriate to operating markets.
- Clear allocation of WhatsApp/Meta and other third-party usage charges.
- Subscription, invoicing, GST/tax process and cancellation policy.
- Support channels, incident process and service expectations.
- Data retention, export and deletion procedures.
- Customer onboarding/offboarding checklist.
- Production backup/restore testing and business-continuity procedure.

22. Recommendation

Build the pilot as a multi-tenant SaaS from day one, but keep the implementation as a modular monolith. Use one shared production platform with strong tenant isolation, allow each customer to onboard its own WhatsApp Business identity, and price the service primarily per venue. Validate the full scan-to-park-to-retrieve journey with one restaurant before investing in advanced automation. After pilot validation, move critical infrastructure to appropriate paid production tiers and introduce standardized onboarding and subscription plans.

23. Proposed Next Steps

- Confirm pilot restaurant and expected daily valet volume.
- Finalize the guest conversation and valet state machine.
- Finalize tenant/venue/user data model and authorization design.
- Create low-fidelity UI wireframes for valet, manager and platform admin.
- Validate current Meta WhatsApp Business Platform onboarding and pricing requirements.
- Build the end-to-end thin slice before secondary features.

- Define pilot KPIs and commercial conversion criteria.